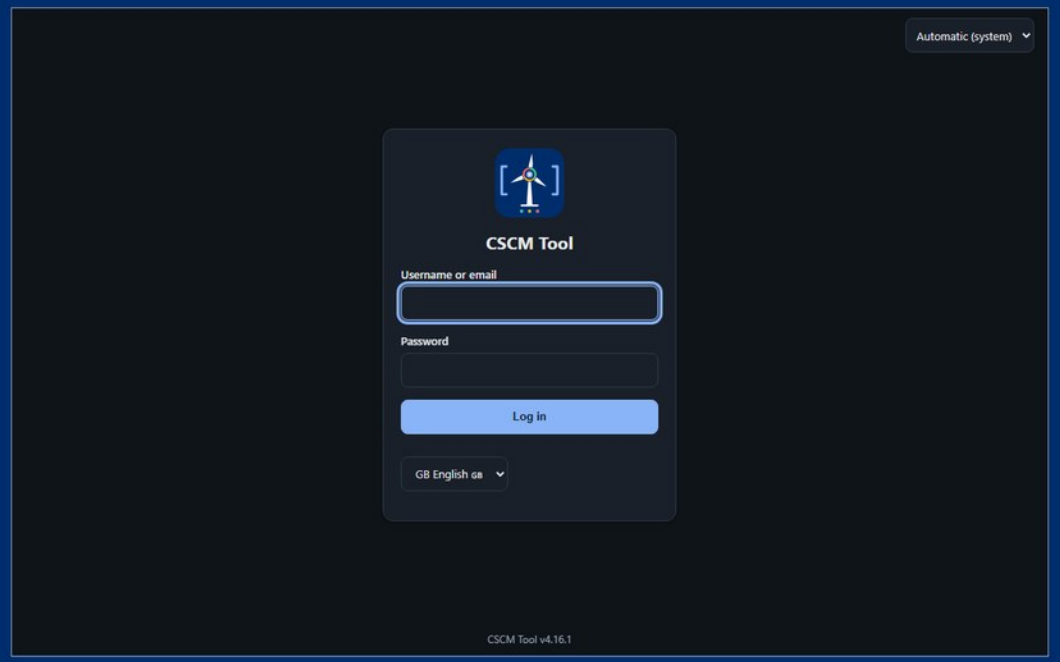
Software Owner and Administrator: Bastian Ritter · bastian.ritter@senvion.in

CSCM Tool

Controller Status Code Management Tool

Design walkthrough for the team



Version 4.16.1

2026-10-27

Main Controllers and Hub Controllers of wind turbines

Design principles

1 Same shell everywhere

Top bar, role-based menu and tokens of the OneTool design system. No CDN, no framework.

2 Colour means something

Roles, lifecycle states and confidence have fixed colours. Every pair is checked for contrast.

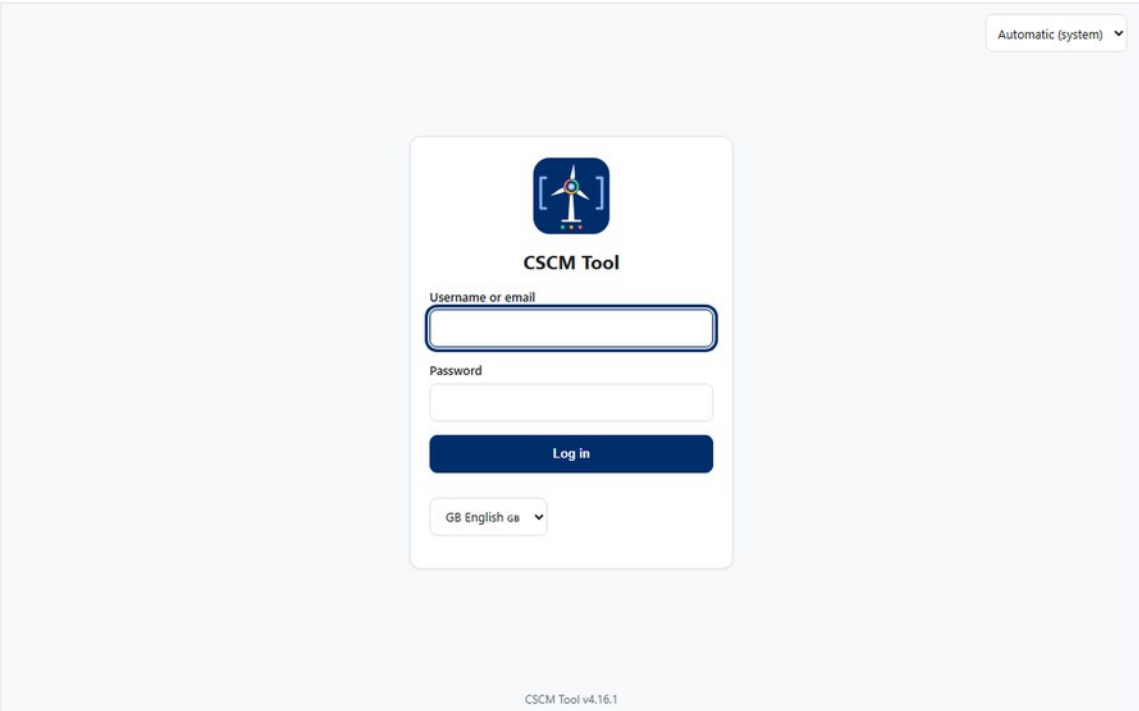
3 Everyone can use it

Keyboard operable, labelled controls, 19 languages, right to left, dark theme.

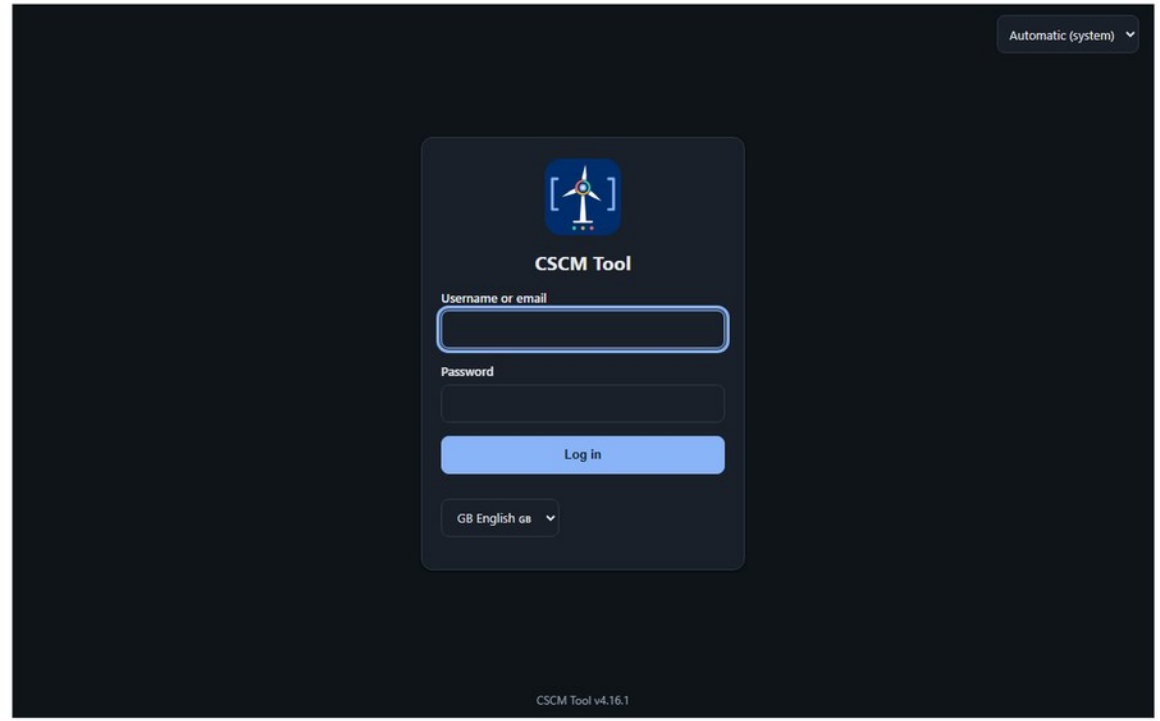
4 Nothing is lost

A rejected form keeps what was typed. Every action is audited.

Login

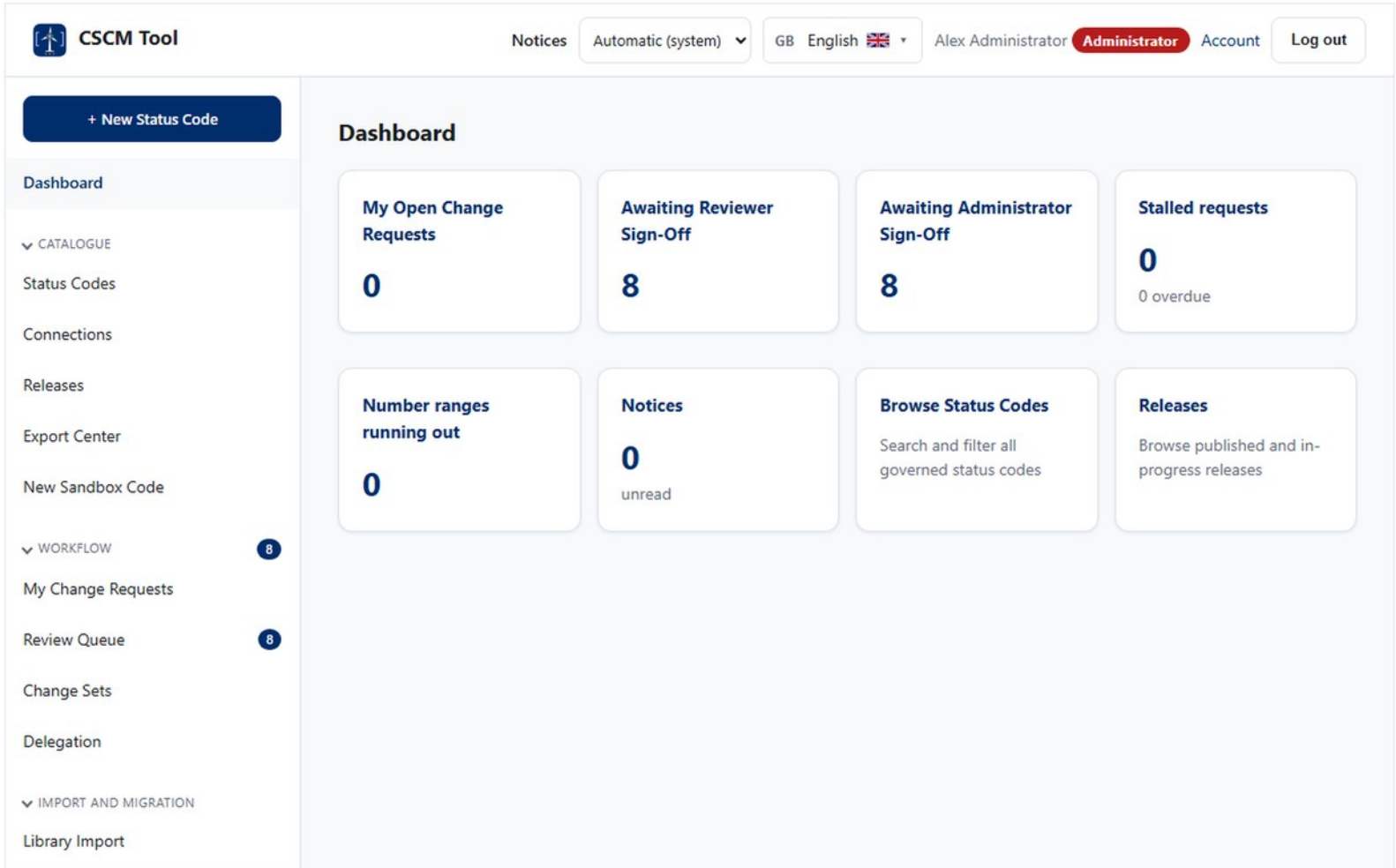


Password sign-in, with the language choice and the theme choice before anybody is signed in.



Where the organisation uses Windows accounts, one button signs in with the Windows identity.

The shell: top bar and menu

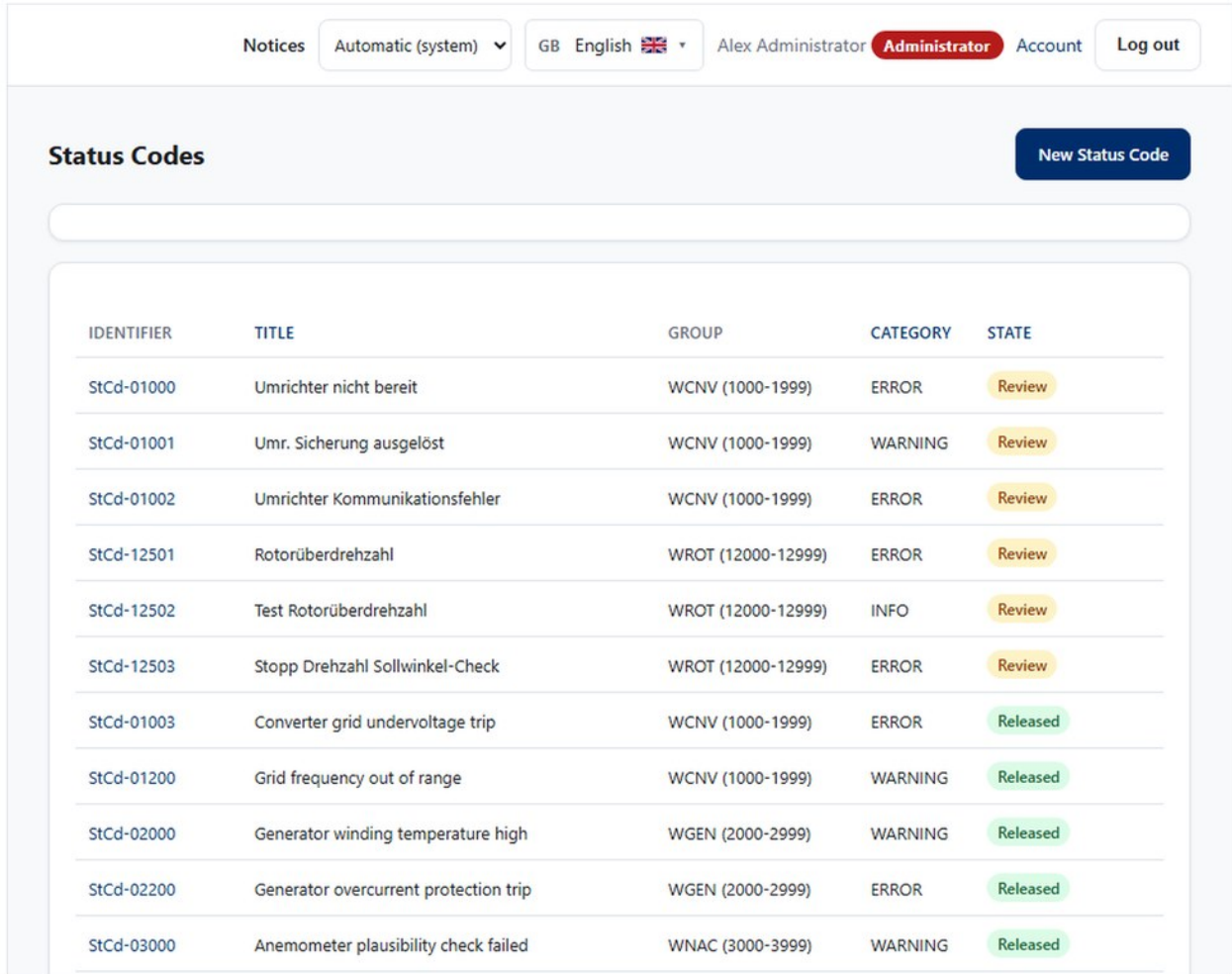


1 **Top bar**
Notices, theme, language, the person and their role badge, account and log out.

2 **Menu**
Groups open and close; the count beside a link is the work waiting there.

3 **Role colours**
Red is the highest role. The colours repeat in lists and reports.

Status Codes: find, filter, share



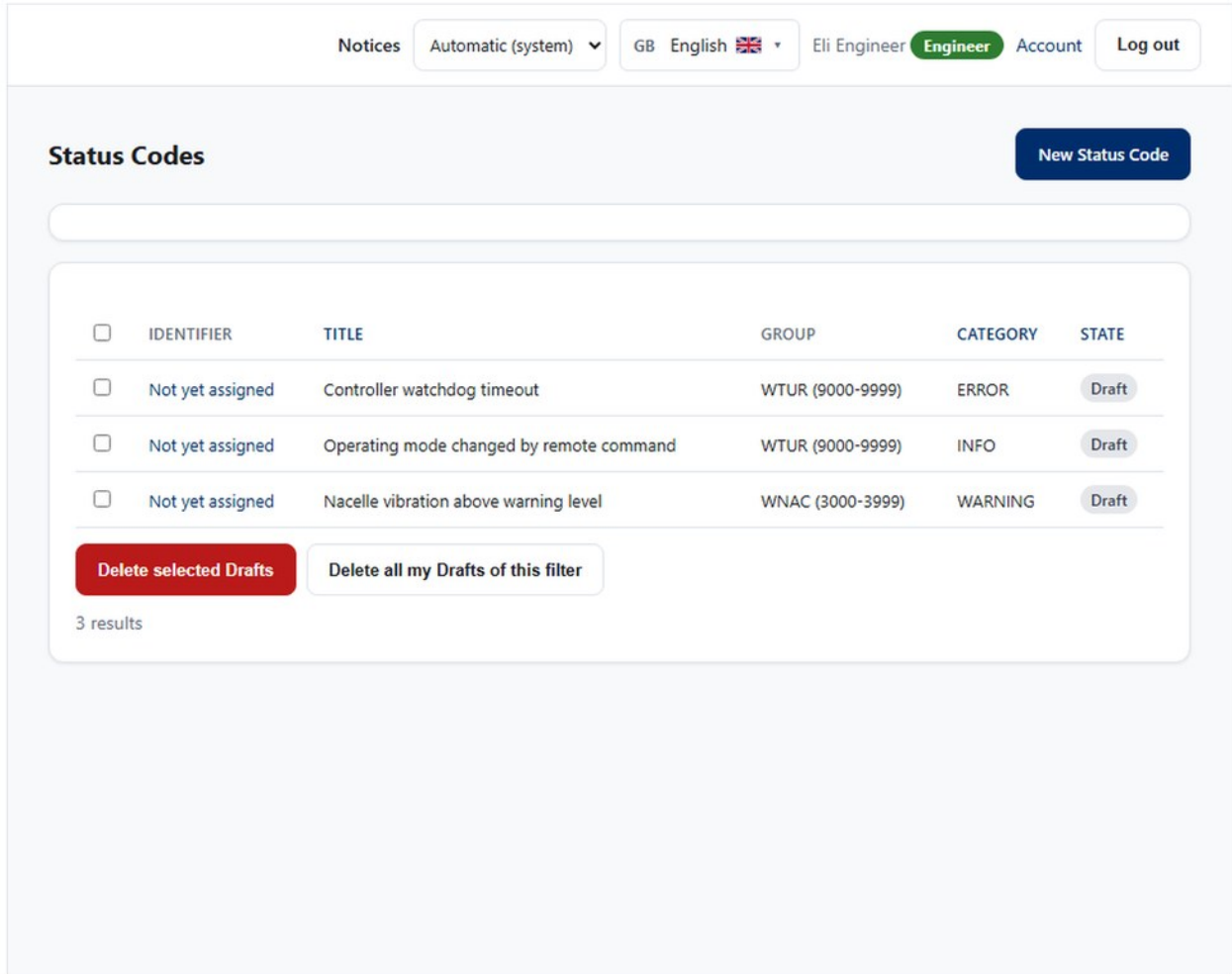
IDENTIFIER	TITLE	GROUP	CATEGORY	STATE
StCd-01000	Umrichter nicht bereit	WCNV (1000-1999)	ERROR	Review
StCd-01001	Umr. Sicherung ausgelöst	WCNV (1000-1999)	WARNING	Review
StCd-01002	Umrichter Kommunikationsfehler	WCNV (1000-1999)	ERROR	Review
StCd-12501	Rotorüberdrehzahl	WROT (12000-12999)	ERROR	Review
StCd-12502	Test Rotorüberdrehzahl	WROT (12000-12999)	INFO	Review
StCd-12503	Stopp Drehzahl Sollwinkel-Check	WROT (12000-12999)	ERROR	Review
StCd-01003	Converter grid undervoltage trip	WCNV (1000-1999)	ERROR	Released
StCd-01200	Grid frequency out of range	WCNV (1000-1999)	WARNING	Released
StCd-02000	Generator winding temperature high	WGEN (2000-2999)	WARNING	Released
StCd-02200	Generator overcurrent protection trip	WGEN (2000-2999)	ERROR	Released
StCd-03000	Anemometer plausibility check failed	WNAC (3000-3999)	WARNING	Released

1. **Filters**
   Search tolerates typos; group, subgroup, platform, lifecycle and category narrow the list.

2. **Columns and saved searches**
   Choose the columns, save a view and share its link.

3. **State badges**
   Draft to Archived, always in the same colours and always with words.

Select, select all, delete Drafts

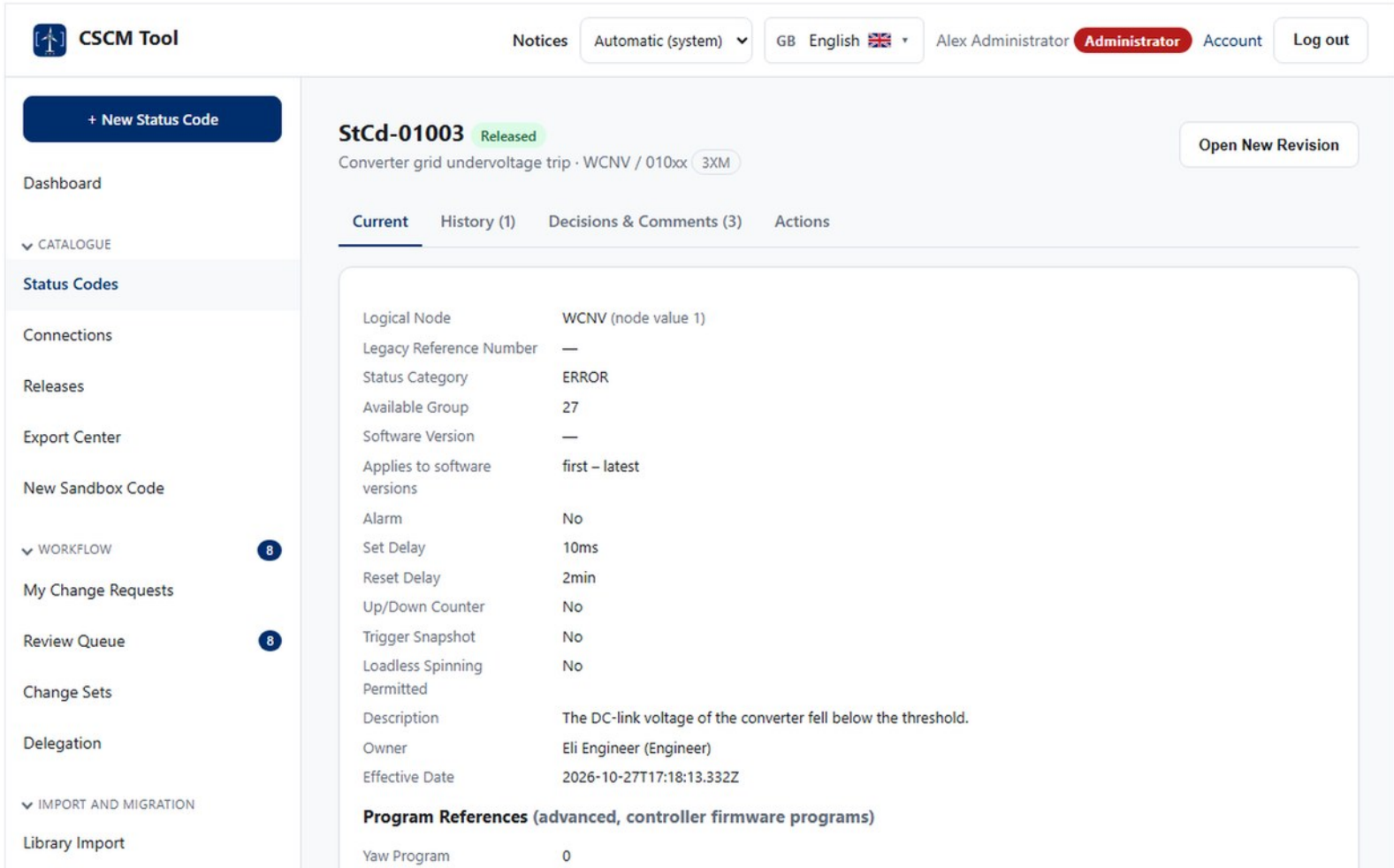


1. **Tick boxes for your own Drafts**
   Only the author can delete a Draft, so only these rows have a box.

2. **Select all**
   The box in the header ticks every Draft on the page.

3. **All Drafts of a filter**
   When filtered to Draft, one button deletes your Drafts on every page.

A status code



1

Identity and state

The number, the state badge and the sign-offs at a glance.

2

Tabs

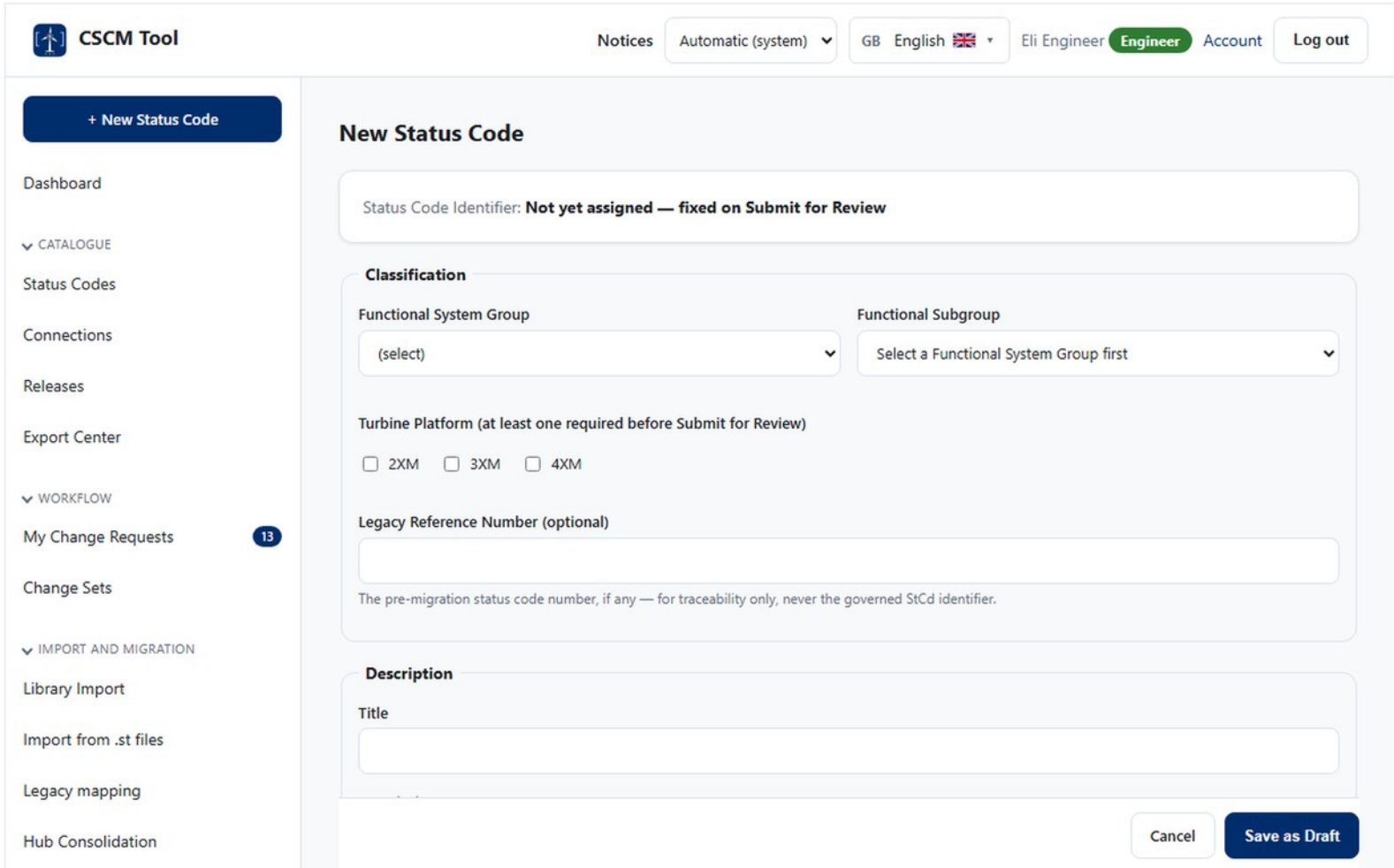
Details, history, comparison, where used, translations.

3

One number, forever

A number is issued once, at submit, and never reused.

Writing a code

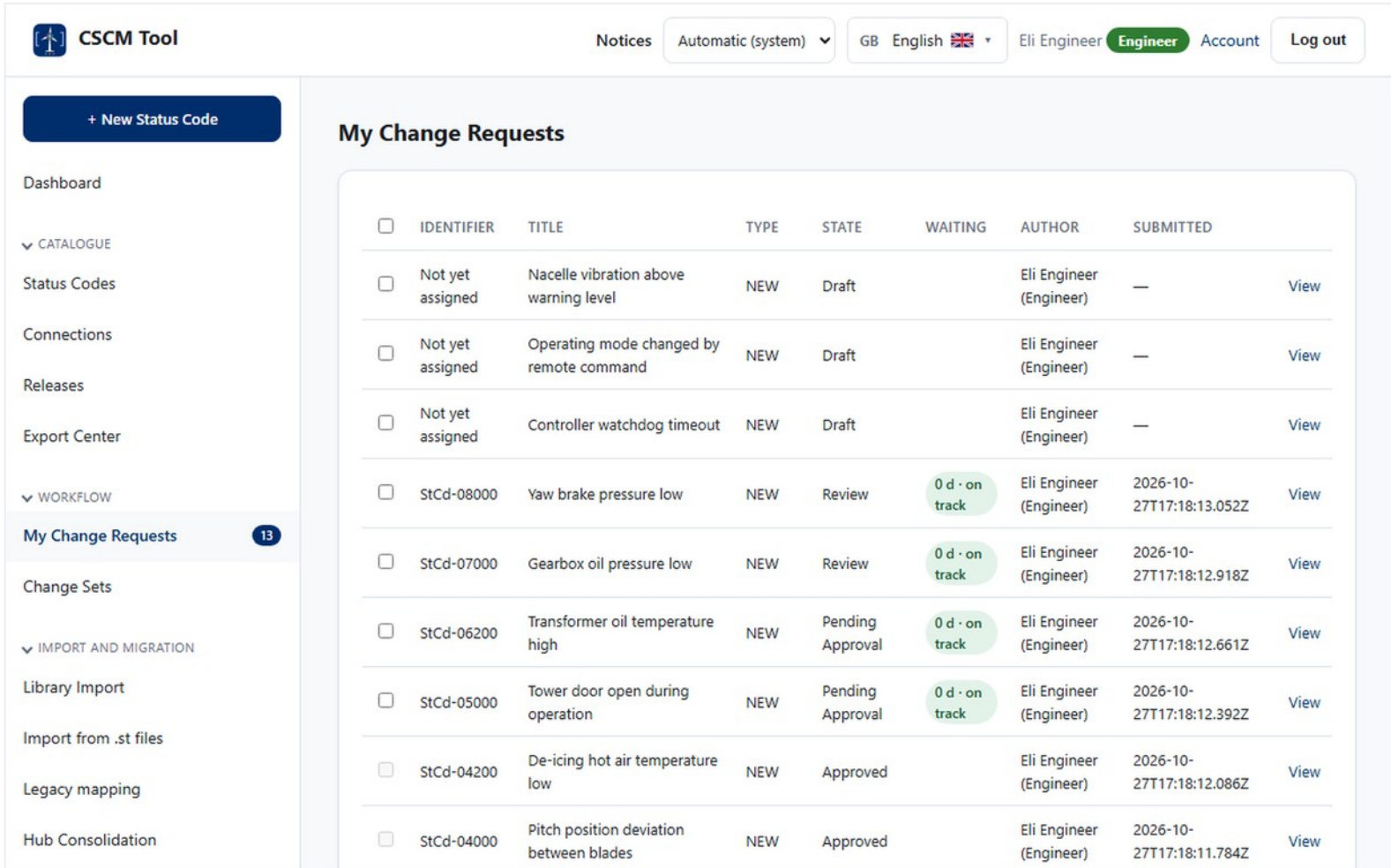


1 **Help beside every field**
Each choice explains itself; subgroups follow the group chosen.

2 **Soft checks**
Duplicates, similar titles and range capacity are shown while typing.

3 **Nothing is lost**
A rejected save keeps every entry and says what to correct.

My work and the queues

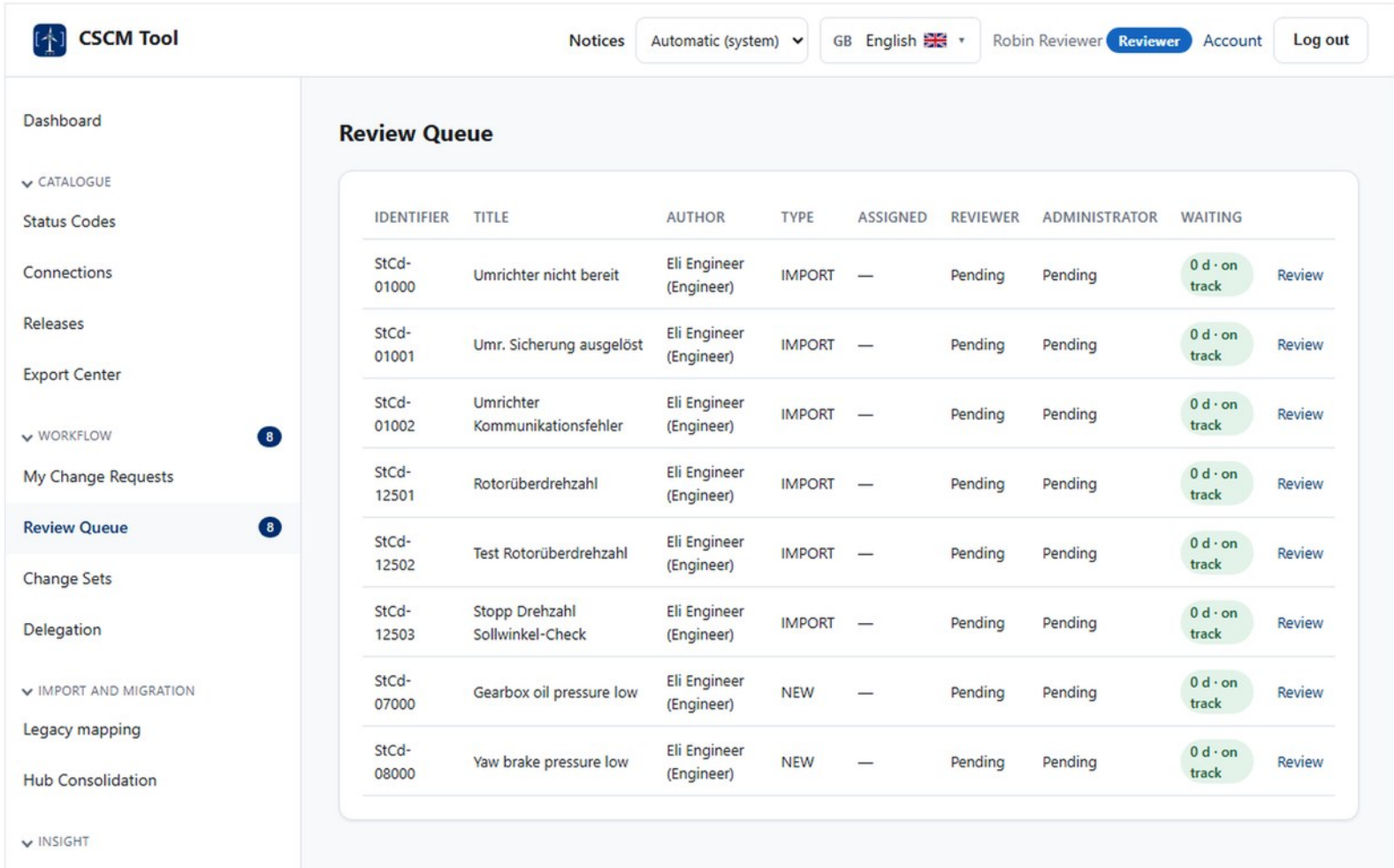


1 **My Change Requests**
Your own requests, their state, and who they wait for.

2 **Select and withdraw**
Tick several and withdraw them at once; each is reported.

3 **Queues for the roles**
Review and approval queues show the age of every request.

Review queue

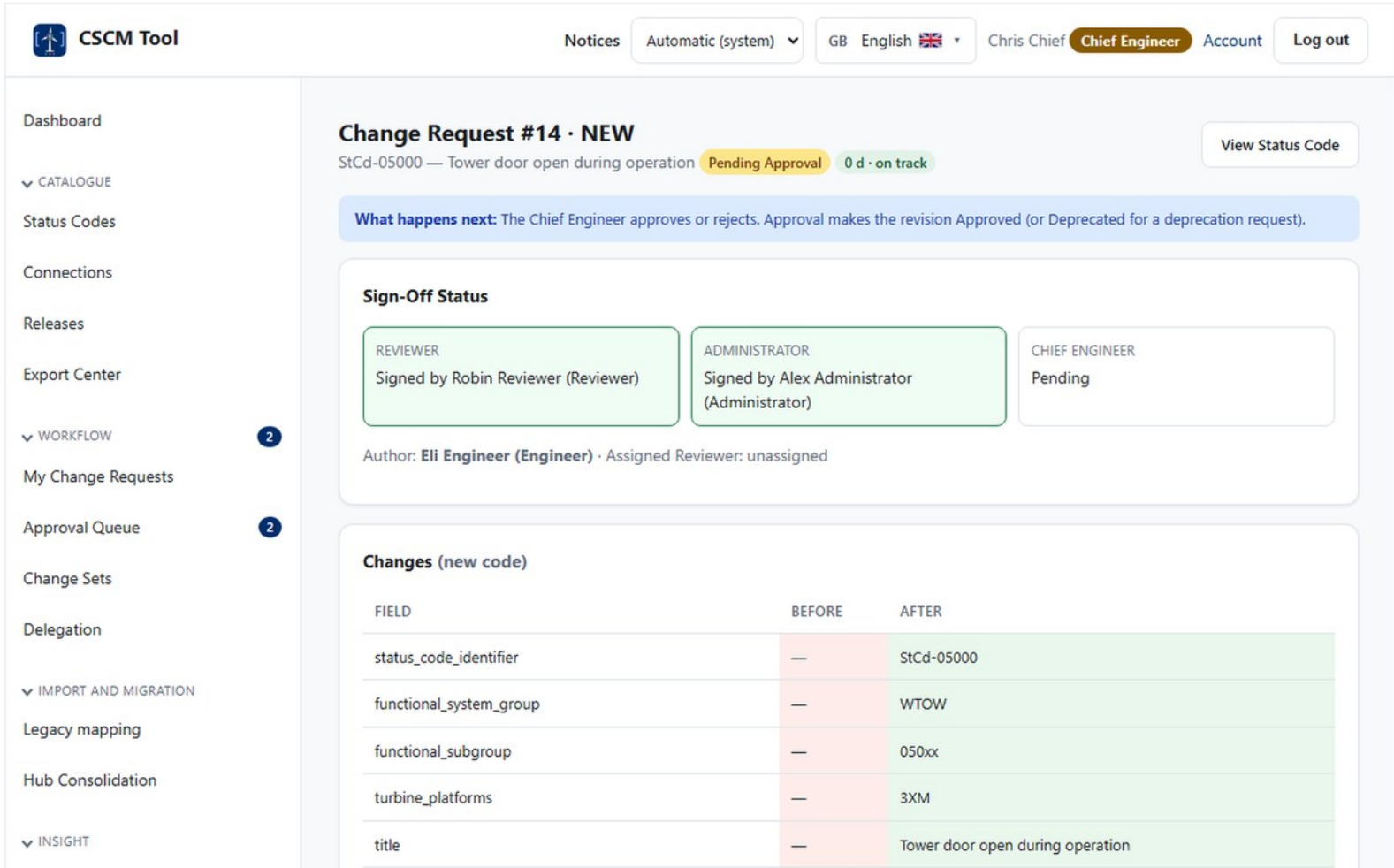


1 **Waiting time as a pill**
Green, amber and red follow the service level of the step.

2 **Open and decide**
One click to the request with its full history.

3 **Separation of duties**
You cannot review your own work.

Sign-off and approval

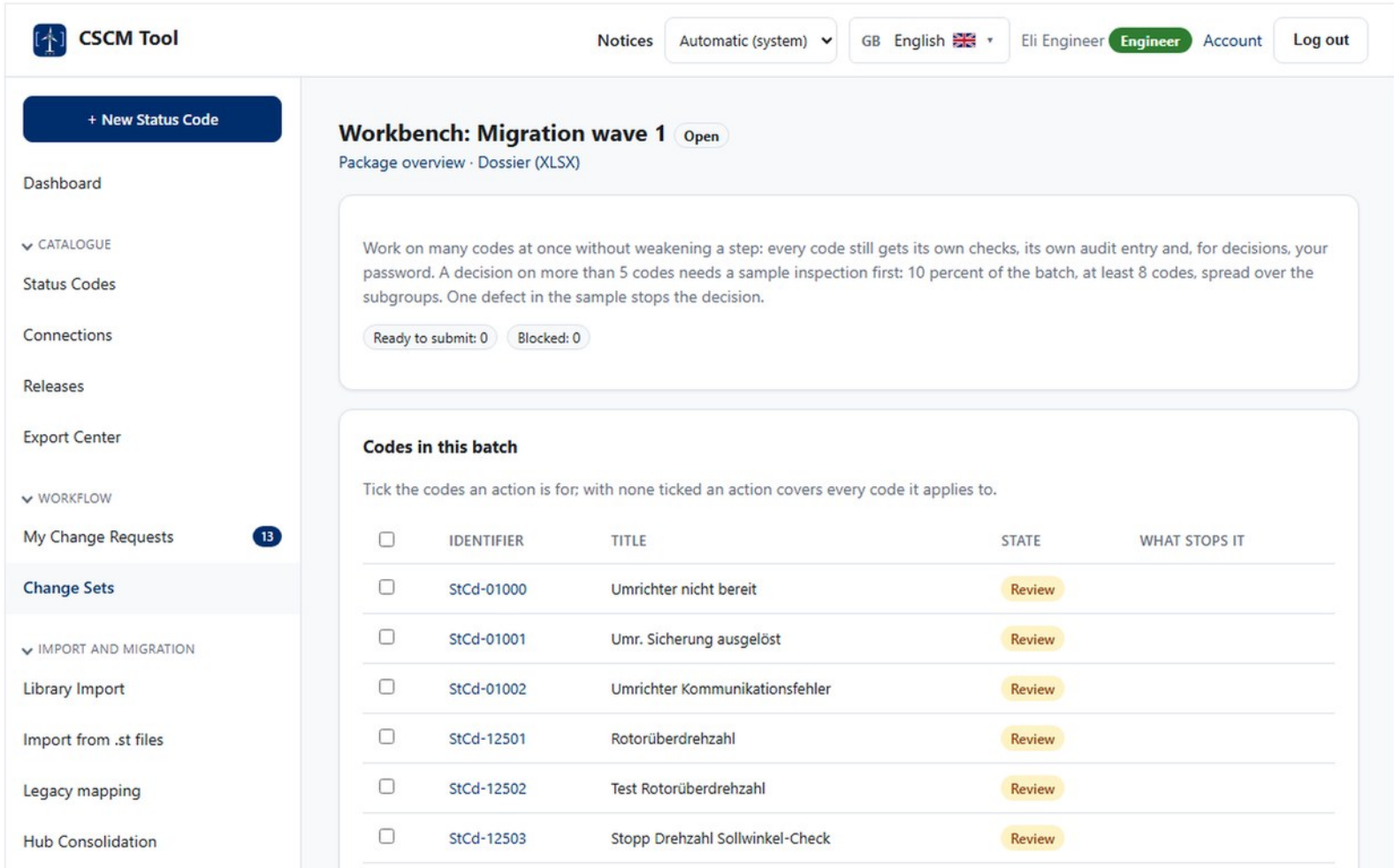


1 **The strip of sign-offs**
Domain, Reviewer, Administrator and Chief Engineer, each with name and time.

2 **Comments and field notes**
Decisions, comments and notes on single fields stay with the request.

3 **Password for decisions**
Sensitive steps ask for the password again.

Change sets and the workbench

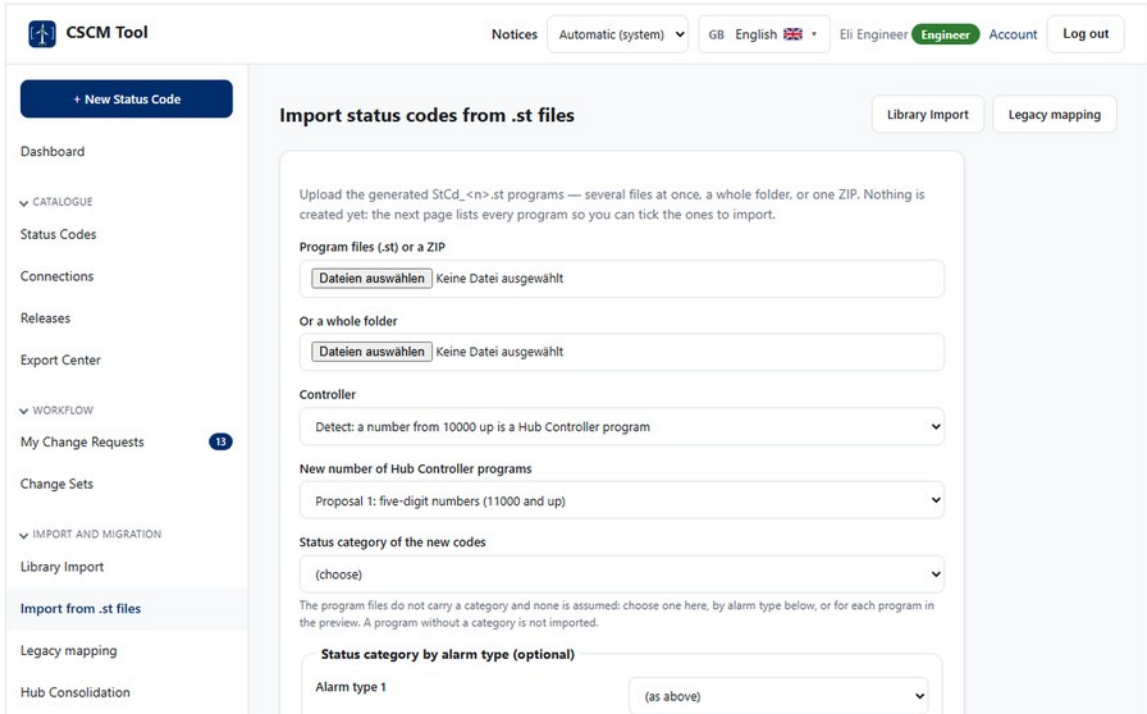


1 **Readiness**
What still stops each Draft from being submitted.

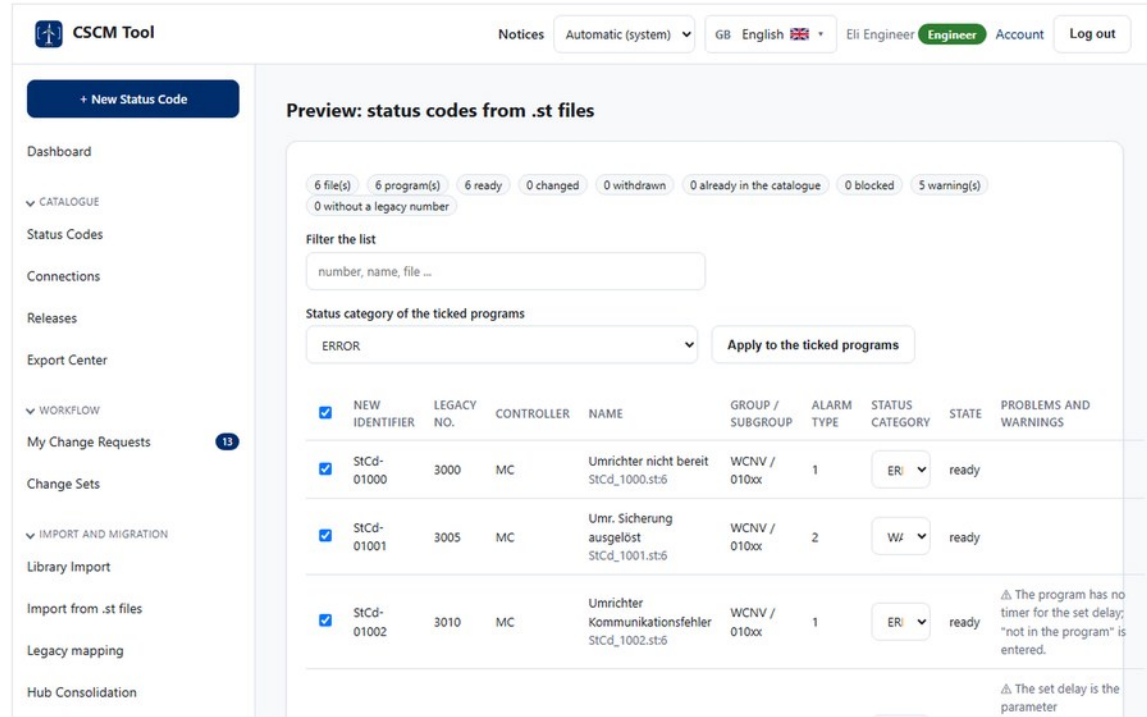
2 **Bulk edit and bulk steps**
Set one value, submit, sign off or decide for the whole batch.

3 **Sample inspection**
Draw a sample, mark each code, and one defect stops the batch.

Importing from controller programs

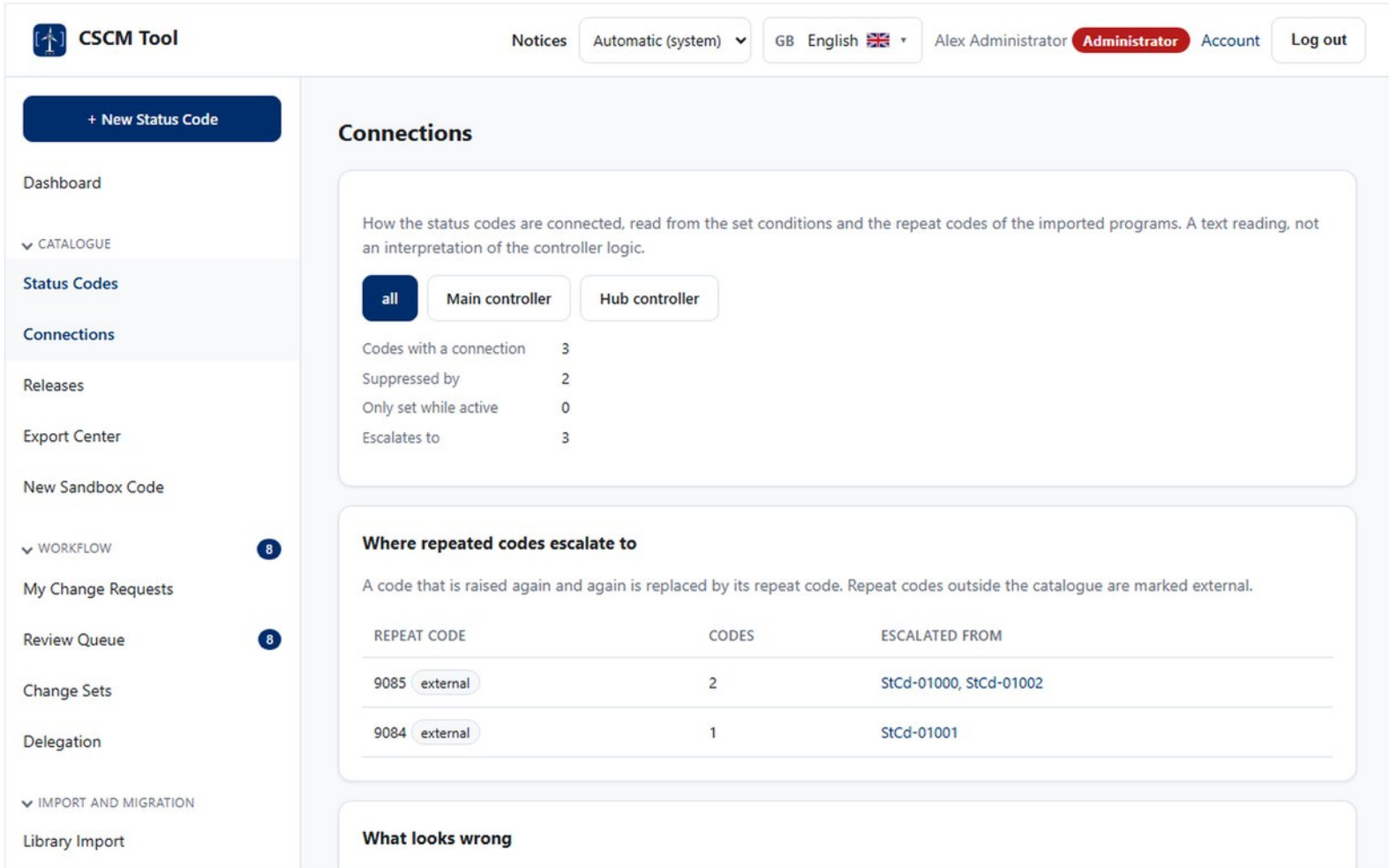


Upload files, a folder or a ZIP. Choose a category for each alarm type.



Every program is a row with its new number, alarm type and a category you can change.

Connections

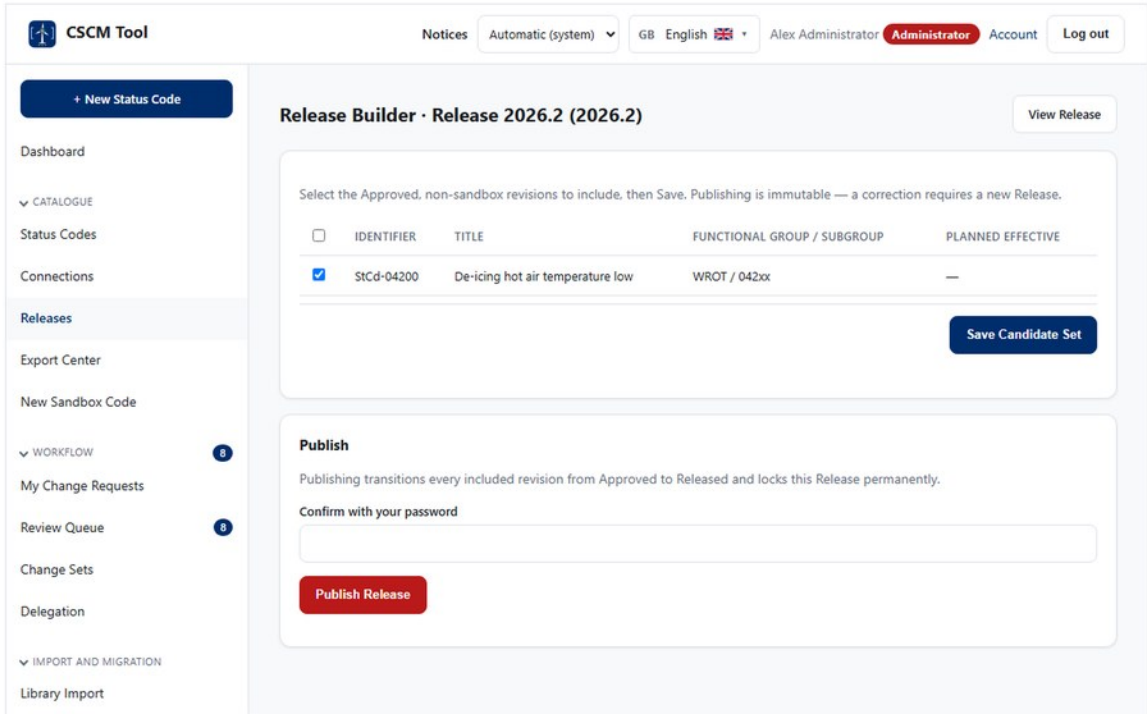


1 **Suppressed by**
A code that is silenced while its cause is active.

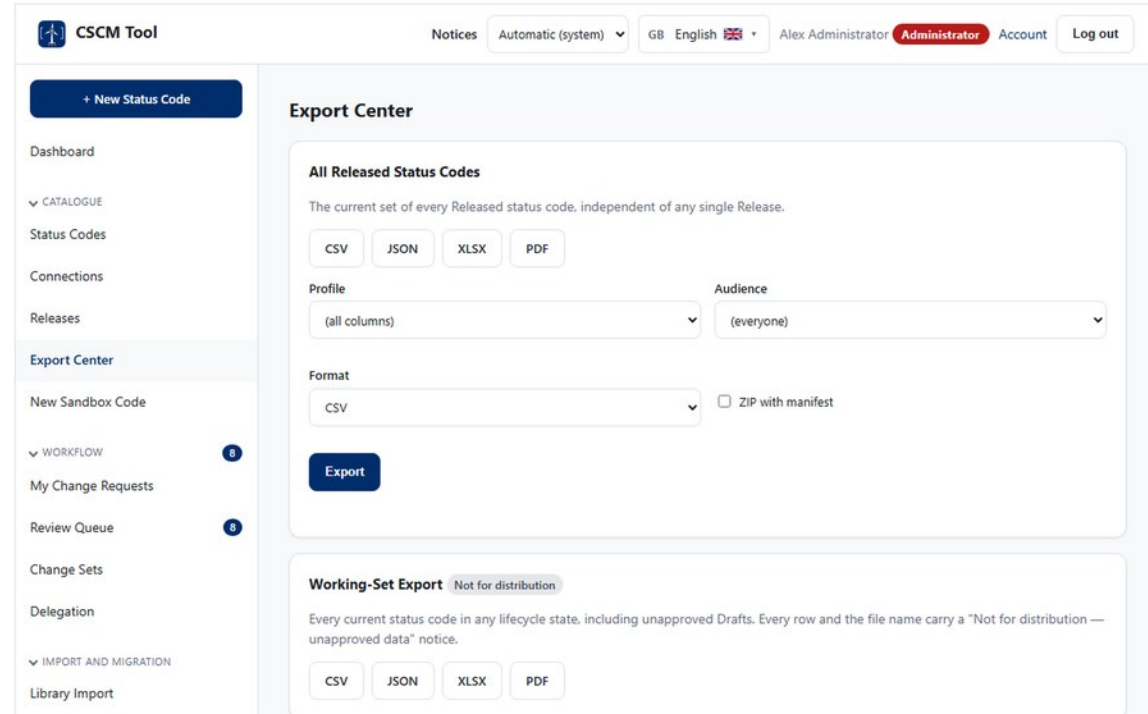
2 **Escalates to**
Where a repeating code is replaced by its repeat code.

3 **Read, not guessed**
Taken from the programs; loops and missing codes are named.

Releases and exports

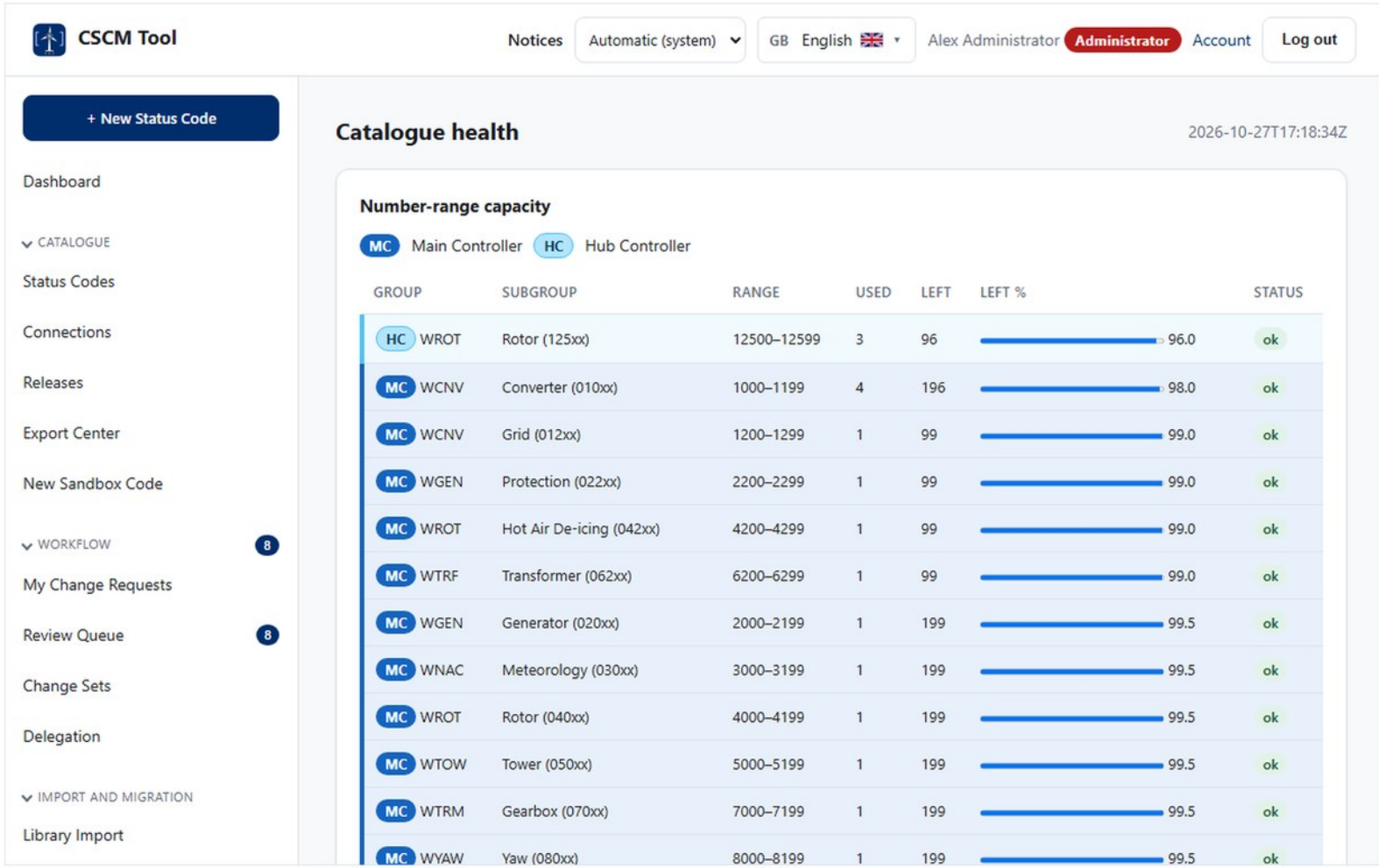


The builder collects approved codes into an immutable release.



Exports and the package, with the download rule applied.

Catalogue health

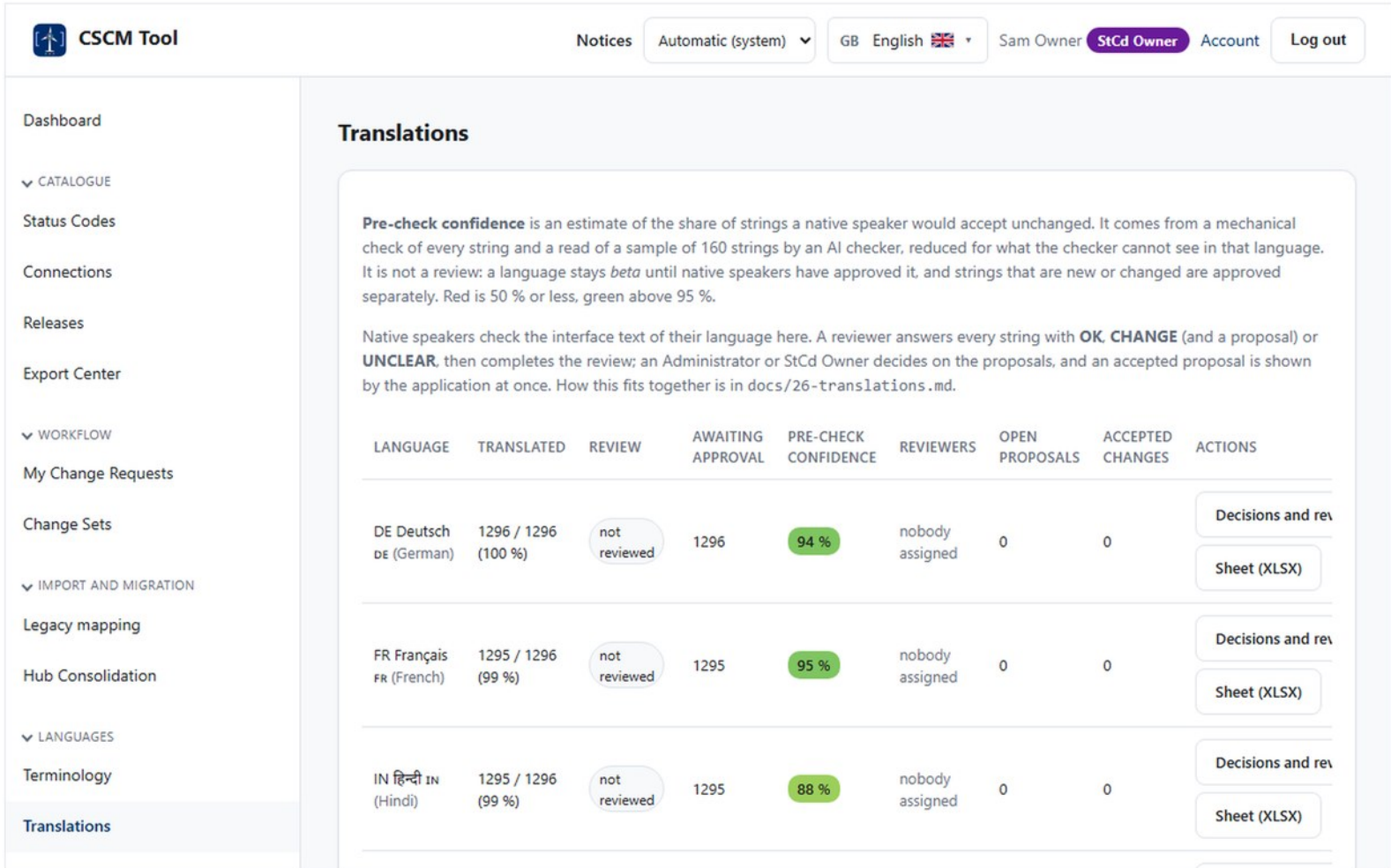


1 **Capacity of the ranges**
How full each number range is, and when it will run out.

2 **Colour for the controller**
Main Controller rows blue, Hub Controller rows light blue.

3 **Stalled work**
Requests that wait too long are listed for the administrator.

Translations and confidence

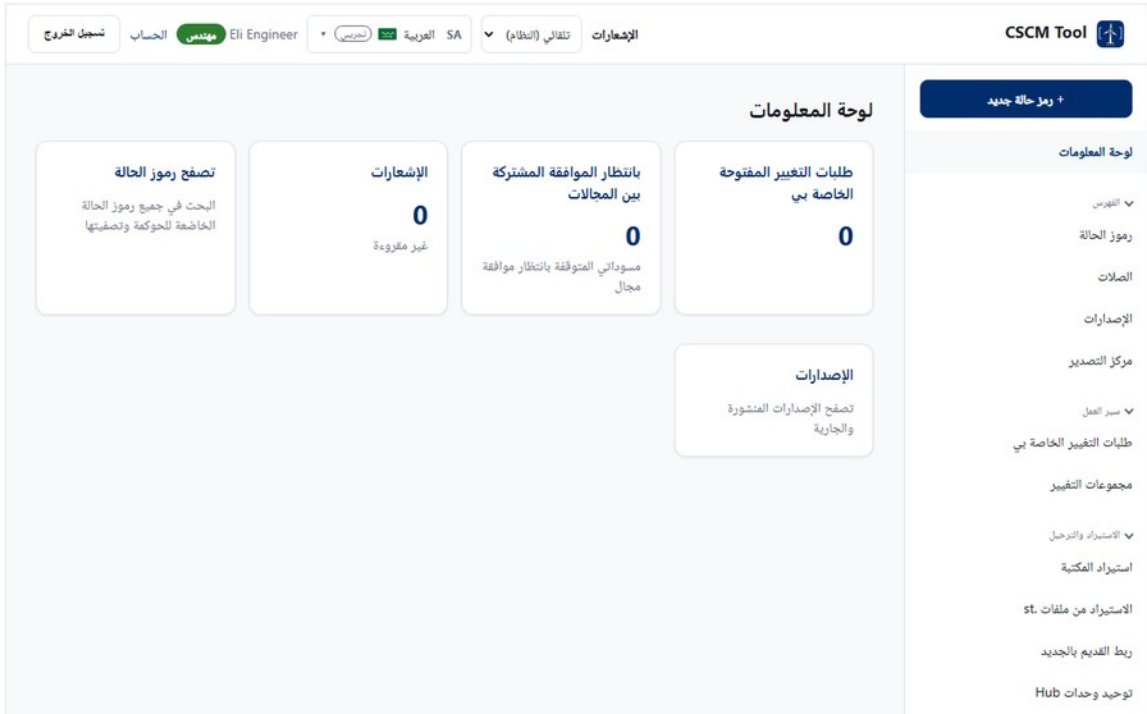


1 **One row per language**
Coverage, native-speaker review and the confidence of the pre-check.

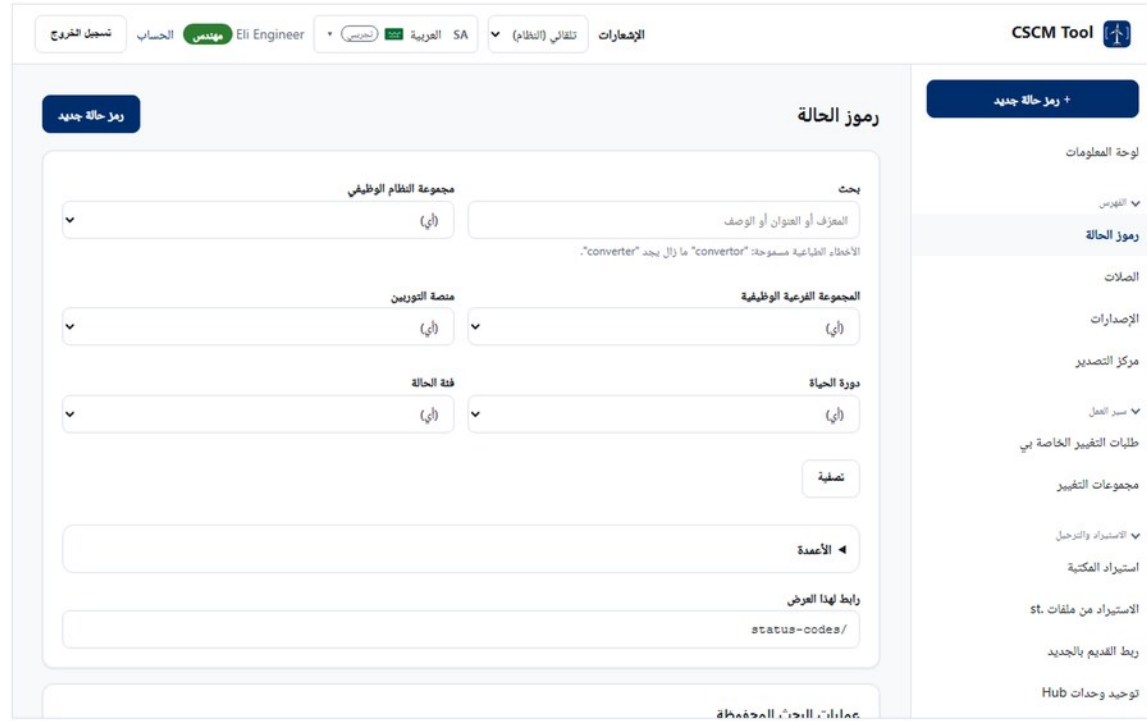
2 **Confidence colours**
Red up to 50 %, green above 95 %.

3 **New words are checked**
Changed or new texts are checked and approved separately.

Languages and right to left

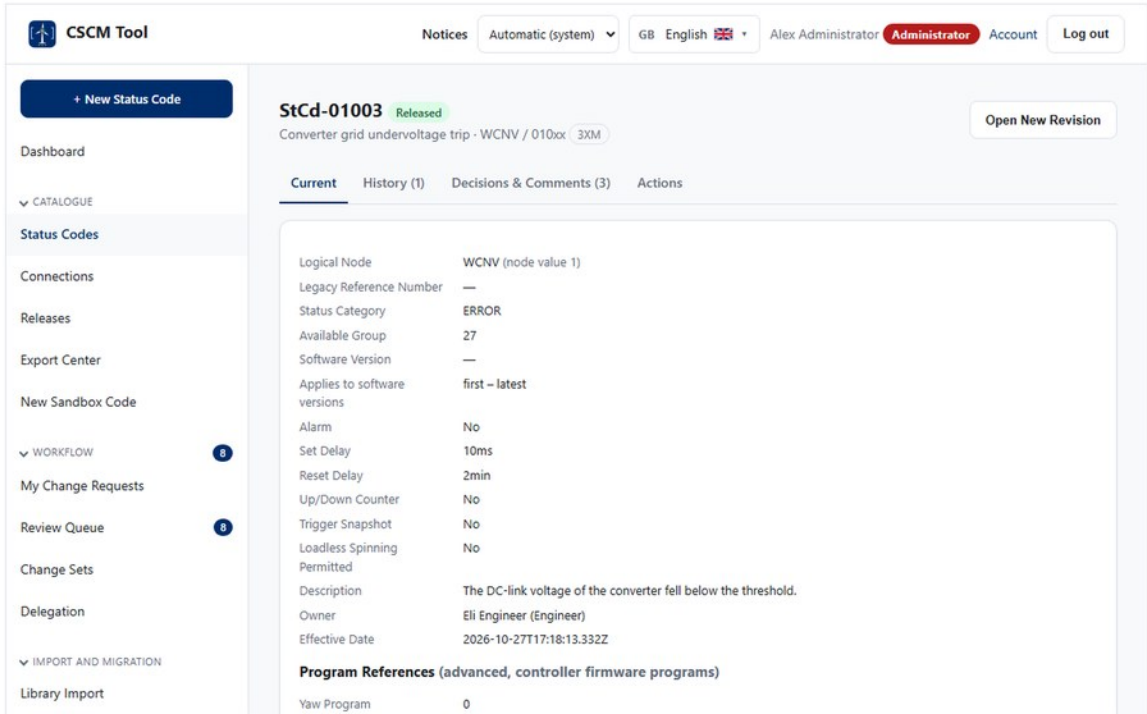


Arabic mirrors the whole layout: menu on the right, tables read from the right.

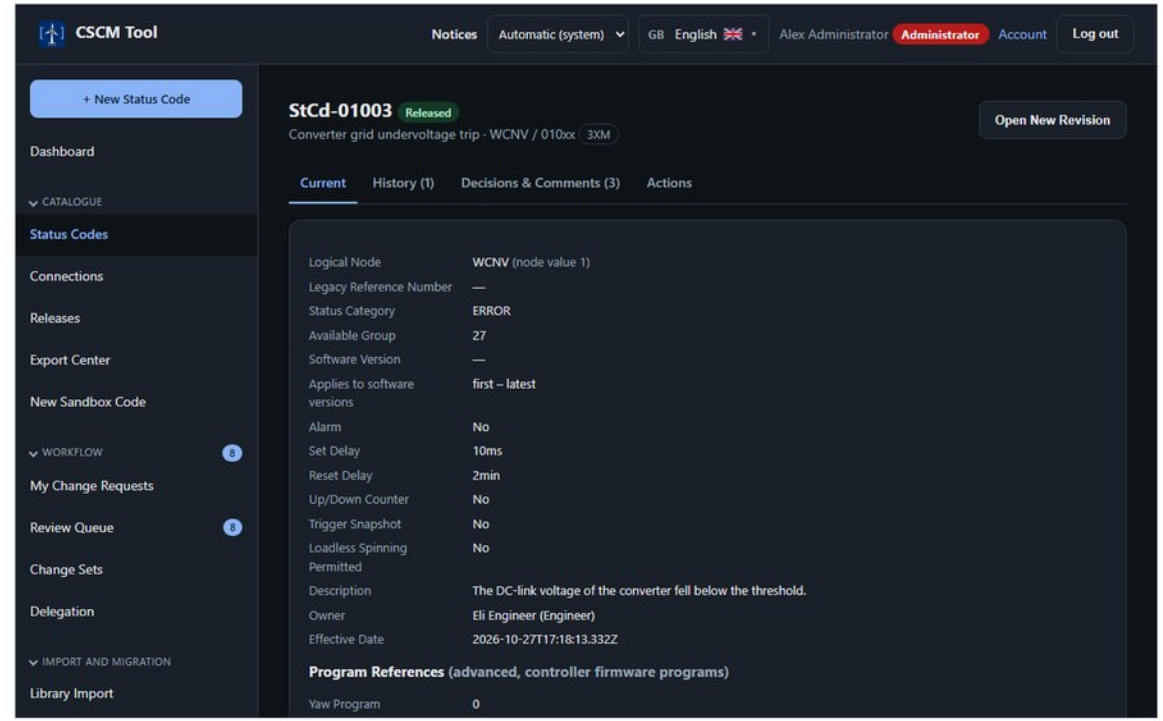


Identifiers and numbers stay left to right inside the text.

Light and dark

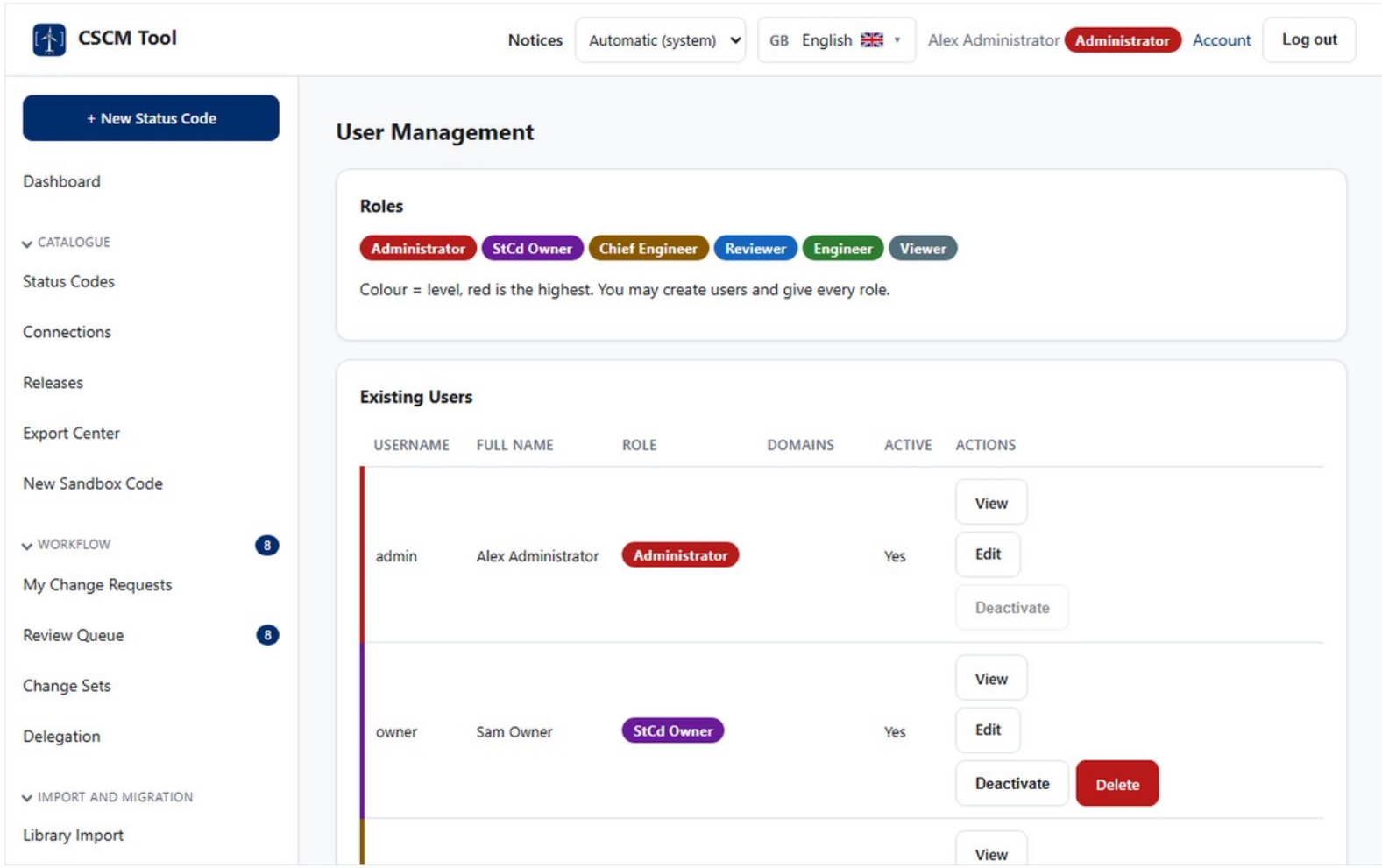


Light: ink on paper, for the office and for print.



Dark: the same page, with every colour pair checked for contrast.

People and roles

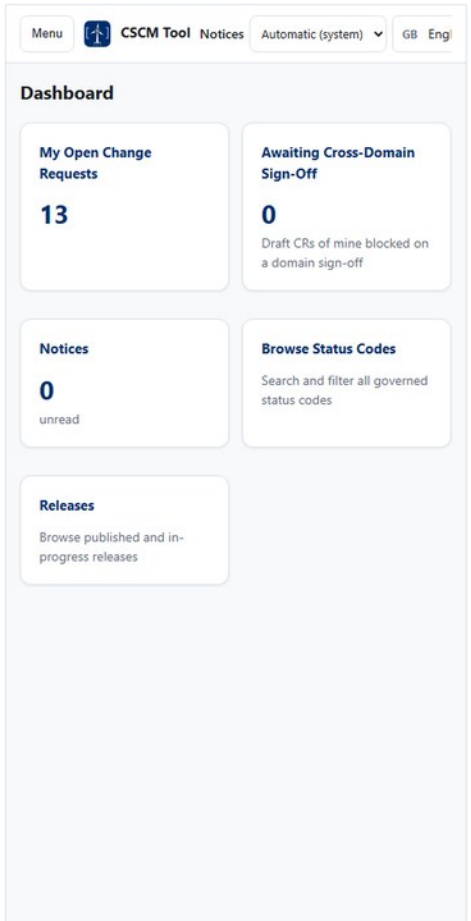


1 **Six roles**
Viewer, Engineer, Reviewer, Chief Engineer, Administrator, StCd Owner.

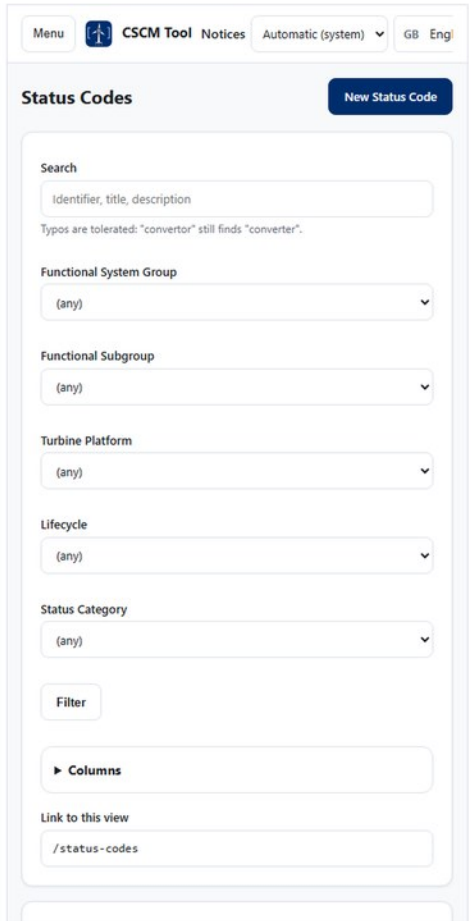
2 **Colour is the level**
The role badge colour is the same everywhere it appears.

3 **Accounts by e-mail**
The e-mail address is fixed and links the Windows identity.

On a phone



Dashboard



Status codes

1 **Menu on demand**
A button opens the menu; Esc closes it.

2 **Touch-sized targets**
Buttons and links are at least 44 by 44 pixels.

3 **No horizontal page scroll**
Wide tables scroll inside their card.

Rules for every new screen

1	Use tokens, never colours	var(--color-surface), var(--tint-warn-bg) … A test fails on a hex colour outside the tokens.
2	Give a dark twin to a new token	A light token without a dark value would stay light on a dark page.
3	Label every control	A visible label or an accessible name; tables use scope on their headers.
4	Words for states, not only colour	Badges carry the state as text.
5	Translate every text	New words go through the confidence check and a separate approval.

The same rules, from login to release

- One shell, tokens for every colour, both themes
- Governance visible in every screen
- Version 4.16.1: 1,035 tests, 19 languages

